Panskura Banamali College(Autonomous)

M.Sc. 2nd Semester Examination -2021 (Online Mode)

Sub: Computer Science Paper: COS-291

Paper Name: DBMS Lab

F.M.: 40 Time : 2 Hours

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020PG009967, paper_code : COS-291 , then file name will be :: 2020PG009967- COS-291.pdf for question 'A' and 2020PG009967- COS-291.mp3 for question 'B']

A. Answer any one question **1×30=30**

1.EMP (Eid,Ename,Salary)
WORKS (Eid,did)
DEPT(did,dname,managerid,floor_name)
Write the following in SQL:
a) Print the name of all employees who work on the 5th floor and earn salary greater than 60,000.
b) Print the name of the DEPT that employee Tina works in.
c) Print the name of all employees who works on floors where koyel works.
d) Give every employee who works in the production dept. a 10% raise in the salary.
e) print names of employee who not work in any dept.

2) Hotel(h_no,name,address)
Room(r_no,h_no,type,charge)
Booking(h_no,g_no,date_from,date_to,r_no)
Guest(g_no,name,address)

a) List all hotel details.
b) List all unoccupied room for the "Grand Hotel"
c) List the name and address of all guest for the "Taj Hotel" on '15-Aug-07'
d) List the guest details currently staying in "Oberoy Hotel".
e) List guest details of the booking 01-Aug-14

3) EMP(e_name, street, city)
WORKS(e_name, company_name, salary)
COMPANY(company_name, city)
MANAGERS(e_name, manager_name)

a) *Find the company with most employees.*
b) *Find the company with the smallest payroll.*
c) *Find all employees who earn more than the average salaryof all employees of their company.*
d) *Find the number of employee for each company.*

e) *Find the name of the employee and corresponding company name who get minimum salary of their respective company*.

4. Consider the following database :
Project (Project#, P_name, Chief-Architect)
Employee (Emp#. E_name)
(a) Get details of all employees in descending order.
(b) Get Emp# of employees working on project no 353.
(c) Get details of employees working on more than two projects.
(d) Get details of employees working on a project named 'Database'.
(e) List the names of the employees who work under the Chief Architect named S. Roy.

5. Consider the following database : Employee (e no, e name, e addr)
Project (p no, p name)
Work no,p no)
(a) Find the number of employees who are working on more than one project.
(b) Print name of the employees who are working on project name 'banking system'.
(c) Print the name of the employees who are not working in any project.
(d) Display all information about the employees whose 2nd character of name is either i', 'or'. 'o'.
(e) Print name and address of the employees with project name with project no. greater than 30.

6. Consider the relational database shown below
Employee(employee_name,street,city)
Works(employee_name,company_name,salary)
Manager (employee_name,manager_name)

(a) Modify the database so that 'Sandip' now lives in "Delhi'.
(b) Give all the employees of 'BOI Corporation' a 10 percent raise.
(c) Give all managers of 'BOI Corporation' a 20 percent raise.
(d) Delete all tuples in the works relation for employees of 'small bank corporation'.
(e) Find the names of the employee whose salary is greater than 50.000.00.

7. Consider the following database :
hotel (h_no, h_name, h_address)
room (r_no, h_no, type, charge)
booking (h_no, g no, date_from, date_to, r_no)
guest (g_no, g_name, g_address)
(a) List all single rooms with a charge below rupees 2000.
(b) List all unoccupled room for the 'Grand hotel'.
(c) List the guest details currently staying in 'Seahawk' hotel.
(d) List the details of all guests staying at the 'Grand hotel from 26th June 2013.
(e) Find the name of the hotel which offers a room with cheapest charge.

8. Consider the following database:
   hotel(h_no,h_name,h_address)
   room(r_no,h_no,type,charge)
   booking(h_no,g_no,date_from,date_to,r_no)
   guest(g_no,g_name,g_address)
   (a) List all single rooms with a charge below rupees 2000.
   (b) List all unoccupied room for the 'Grand hotel'.
   (c) List the guest details currently staying in 'Seahawk' hotel.
   (d) List the detals of all guests staying at the "Grand' hotel from 26th June 2013.
   (e) Find the name of the hotel which offers a room with cheapest charge

Marks distribution:

a. Problem Definition -	10 Marks
b. Algorithm/Flow Chart/ Pseudo-code–	15 Marks
c. Program listing -	03 Marks
d. Output/Result with discussion-	02 Marks

B. Answer any five question **5*2=10**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What are the different types of languages that are available in the DBMS?
2. What is the purpose of SQL?
3. What are some examples of open source and commercial Relational DBMSs?
4. What is a database model? and name a few common database models?
5. What are the different levels of data abstraction?
6. What is normalization and de-normalization?
7. What are the main differences between Primary key and Unique Key?
8. Explain Entity, Entity Type, and Entity Set in DBMS?
9. What are the different levels of abstraction in the DBMS?
10. What are different types of joins in SQL?

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